

HGI Door Builder inventory architecture audit

Date: October 8, 2026. Repository baseline: 40abeeb924355dd78d3679a2d10a1496084a6ba0.

1. Scope and current architecture

This is an inspection and architecture report only. No runtime code, inventory, product rules, assets, configuration UX, Visualizer, PDF, lead integration, or production data flow was changed. No database schema or migration was implemented. FeneVision is not part of this proposal.

The application currently has a **build-time product-choice catalog**, not a live inventory service. TypeScript arrays, maps, allowlists, filename conventions, and functions determine the selectable products. There is no inventory API/database read in the customer configuration path. Existing Supabase integrations serve dealers, leads, and AI usage, not product inventory.

The effective catalog is assembled in layers:

- Hardcoded line/style, finish, glass, hardware, and sidelite records
- > derived catalog records and scoped asset maps
- > material/style/finish/glass/grid availability filtering
- > customer visibility and grouping
- > display sorting and normal configuration state
- > DoorPreview, complete entrance capture, PDF, submission, AI resolution

src/App.tsx is both the configuration orchestrator and a substantial source of inline catalog policy. Moving only src/data into Supabase would therefore NOT reproduce current availability.

Four responsibilities to keep separate

Responsibility	Current examples	Future location
Product data	Names, codes, colors, materials, grains, glass designs, hardware styles	Supabase catalog tables
Commercial compatibility/availability	Style-material membership, glass allowed for an opening, sidelite family membership, allowed grid combinations	Explicit catalog relationships; application evaluates them
Asset mapping	Slab artwork, overlays, masks, thumbnails, handing/view hardware images	Asset records and typed bindings; renderer remains in code
UI and rendering behavior	Steps, sorting, selected state, default/fallback selection, geometry, handing transforms, glass/frame alignment	Application code, NOT database UI scripts

There is no discovered authoritative source for on-hand quantities, warehouse locations, prices, lead times, manufacturer discontinuation dates, or customer-selectable manufacturing width/height. “Available” currently means allowed by the builder’s tables/rules and represented by its assets; it does not prove commercial stock or manufacturer availability. HGI must define whether the later inventory project also needs stock management.

2. Inventory sources and files

Paths below are repository-relative. H = authored/hardcoded; D = derived. Flags describe the module’s role, not an assertion that every field has that property. This registry covers the product data modules and the inline/consumer rules that affect the effective catalog.

File	Objects/functions and format	H/D	Assets	Rules	Labels / IDs or codes	Derivation and consumers
src/data/productCatalog.ts	signatureStyles, steel20, steel22, brushed, textured: code name string arrays parsed by parseStyles; productCatalog: DoorLine[]; doorTypeOptions, doorLineChoices; glassDoorCodes: Set	H + D	Line images and style thumbnail lookup	Grain/color/paint -only policy and style membership	Both	Builds styleRecords, catalogDoorStyles, variants; exports doorLineChoicesForStyle, stylesForDoorType, doorStyleSupportsGlass, autoGrainForDoorLine, finishTypesForDoorLine, finishesForStyle, resolveDoorProduct; consumed by App, draft validation, AI/server and renderers
src/data/options.ts	doorStyles alias; toFinish, finishes: Finish[]; hardware re-export	D from catalog and palettes	Constructed finish swatch paths	Finish type metadata, not a separate availability catalog	Both	Shared entry point for App, renderers, draft and server
src/data/finishes.ts	paintColors, stainColors: ID/name/hex/ProMatch arrays; cladColors: restricted paint subset	H + D	Indirect through options.ts	Clad palette restriction	Both	Supplies slab, jamb, glass-frame colors and finish swatches
src/data/glassOptions.ts	glassPreviewAssets, glassThumbnailOptions, variantThumbnailOptions; opening-specific *Overlays maps; glassOverlayAssetUrl; glassOptions: GlassOption[]	H + D	Thumbnails + per-opening overlays	Approval filtering and opening membership encoded through overlay presence	Both	Main glass catalog; reused by CR14SL, App, draft and server/renderers

File	Objects/functions and format	H/D	Assets	Rules	Labels / IDs or codes	Derivation and consumers
src/data/decorativeGlass.ts	approvedDecorativeGlassNames, approvedDecorativeGlassIds, knownDecorativeGlassIds and approval helpers	H	Indirect	Decorative allowlists	Both	Filters main and sidelite glass source tables
src/data/privacyGlass.ts	approvedPrivacyGlassNames, approvedPrivacyGlassIds, knownPrivacyGlassIds and approval helpers	H	Indirect	Privacy allowlists	Both	Filters glass and helps App infer main glass family
src/data/customerGlassAvailability.ts	isCustomerSelectableGlass	H predicate	No	Hides Linen by ID/name while retaining records	ID + label matching	Applied to visible main/sidelite choices; not deletion of legacy records
src/data/fslGlass.ts	allFslGlassOptions -> fslGlassOptions; fslGlassCategories; fslStandardFlatRules, fslLowEFlatRules, fslStandardStyleRules, fslLowEStyleRules, fslExternalPatterns, fslSdlPatterns	H + D	Generated FSL glass/grid paths; Prairie/Arts & Crafts maps	Category approvals, coating/style/pattern/color/width combinations	Both	App's sideliteGlassCatalogs, draft validation, server and rendering
src/data/f48slGlass.ts	allF48slGlassOptions -> f48slGlassOptions; f48slGlassCategories, f48slStandardStyleRules, f48slLowEStyleRules, grid/Prairie/Arts & Crafts resolvers	H + D	F48SL-specific paths and shared grid art	Scoped glass and grid availability	Both	Same consumers as FSL
src/data/sslGlass.ts	allSslGlassOptions -> sslGlassOptions; sslGlassCategories, sslStandardStyleRules, sslLowEStyleRules, grid/Prairie/Arts & Crafts resolvers	H + D	SSL paths and shared grid art	Scoped glass and grid availability	Both	Same consumers as FSL
src/data/s2slGlass.ts	s2slGlassOptions, s2slGlassCategories, s2slStyleRules	H	Clear glass asset	Only one glass row; grid rule map empty	Both	Scoped S2SL catalog
src/data/cr14slGlass.ts	cr14slGlassOptions, cr14slGlassCategories, cr14slStyleRules, cr14slGridAsset	D from main CR14 overlays and approvals	Reuses main CR14 glass art	Category filters; empty grid behavior reused from S2SL	Both	Scoped CR14SL catalog; no populated grid catalog

File	Objects/functions and format	H/D	Assets	Rules	Labels / IDs or codes	Derivation and consumers
src/data/sideliteAssets.ts	SIDELITE_SLABS, SIDELITE_GLASS_MASKS; sideliteAssetFamilyForSlab, sideliteStylesForFamily, sideliteSlabAsset, sideliteGlassMask	H	Family/style -> slab/mask paths	Material/grain -> allowed sidelite styles	IDs/codes; UI labels elsewhere	App uses available asset-map keys as selectable membership; preview/PDF/AI reuse resolver
src/data/sideliteConfigurations.ts	sideliteProductOptions, singleDoorBuilderOptions, doubleDoorBuilderOptions; codeByInput, normalizeLegacySidelite, resolveSidelitePosition, sideliteBuilderOptions, sideliteProductCode/Label	H + D	No product raster mapping	Semantic hinge/lock relationship ↔ physical placement	Both	Shared preview, saved draft, AI and submission translation
src/data/hardwareAssets.ts	hardwareAssets: HardwareAsset[], resolveHardwareAsset	H	Baldwin source artwork	Handing/view lookup	Manufacturer/style/finish labels	Expanded/filtered by hardware.ts
src/data/schlageHardware.ts	schlageHardware, resolveSchlageHardware	H	Card, exterior and interior art	Style/finish lookup	Both through derived IDs	Expanded for handing/view in hardware.ts
src/data/hardware.ts	finishColors, schlageAssets, allHardwareAssets, hardwareOptions, resolveHardwareOption, hardwareDisplayName; finishSwatchAssets, hardwarePreviewAssets, asset URL resolvers	H + D	Multiple product/card/view/finish art layers	Preferred picker handing; asset fallback; knob-only crop metadata	Both	App hardware groups; DoorPreview, PDF, draft, AI; IDs slug manufacturer/style/finish/handing
src/data/doorConfigurationRules.ts	doorConfigurationRules, doubleDoorLockPrepOptions, savannahRequiredProductOption; leaf count and doorHardwarePlacements functions	H	Entry/lock-prep artwork indirectly	Single/French/Savannah structure, active leaf, DDLLBO/DLLAC/DDLLKP, HINGEOJ	Both	Configuration, preview, capture, PDF, AI/submission consumers
src/data/doorHanding.ts	doorHandingSides and handing-side mapping	H algorithm	No	Exterior/interior hinge and lock orientation	Swing codes; UI labels in App	Shared hardware/sidelite placement

File	Objects/functions and format	H/D	Assets	Rules	Labels / IDs or codes	Derivation and consumers
src/data/doorPreviewAssets.ts	smoothPaintDoorPreviewAssets, exactDoorLinePreviewAssets, texturedPaintDoorPreviewAssets, four signature*PaintDoorPreviewAssets; doorPreviewAssets, resolveDoorPreviewCandidates, resolveDoorPreviewAsset, finishTypesForPreviewAssets	H + D	Slab artwork and fallback candidates	Asset availability; smooth-vs-grained safeguards	Style/line codes	Preview, server, tests and PDF; map presence can gate finish availability
src/data/doorStyleThumbnailAssets.ts	Thumbnail map and fallback; getDoorStyleThumbnailAsset, hasMappedDoorStyleThumbnailAsset	H	Door-style card art	Code-based resolution	Codes	Imported while derived style records are built
src/data/glassMaskAssets.ts	Opening-code mask map and resolver	H	Main glass masks	Code -> opening mask	Codes	Preview/PDF/reference rendering
src/data/glassFrameMasks.ts	Shape maps, separateOpeningCodes, frame-mask creation helpers	H + D algorithm	Derived masks	Oval/round/eyebrow/diamond vs rectangle; separate-lite geometry	Codes, not picker labels	Renderer geometry, not stock data
src/data/heroPresets.ts	Curated HERO_PRESETS/HeroPreset configurations; heroDoorFilename	H	Generated hero art	Marketing presets, not commercial availability	Preset/product keys	App hero and scripts/generateHeroDoorImages.ts
src/App.tsx	doorConfigurationOptions, sideliteStyleOptions, signatureGrainChoices, grainThumbnails, doorSwingOptions, glassCategoryChoices, retroGlassCategory, hardwareStyleGroups; clearGlassIds, three legacy*GlassIds, f48GlassOptionIds, glassCategory	H + D	Option images and group cards	Main glass filtering, family classification, defaults, visible controls	Both	Actual customer-visible effective catalog; source module arrays alone are insufficient
src/App.tsx	gridLocations, gridStyles, lowEGridStyles, flatGridPatterns, allGridPatterns, internalGridPatternCodes, three color-code maps, *GridAsset, *GridRules, *PrairieRules, gridRuleWidths; sideliteGlassCatalogs	H + D	Static + constructed grid paths	F/F48/S and sidelite coating/location/style/pattern/color/width availability	Both	Builds compatibleGridPatterns/Colors/Widths, previews and GridConfiguration

File	Objects/functions and format	H/D	Assets	Rules	Labels / IDs or codes	Derivation and consumers
src/App.tsx	availableFinishes, effectiveFinishTypes, jambFinishOptions, glassFrameFinishOptions, matchedGlassFrameFinish, visibleSideliteStyleOptions, availableGlass, derived pages and selection effects	D	Calls catalog resolvers	Cross-selection exclusions and clearing invalid downstream selections	Derived labels + IDs	UI orchestration; retain algorithms rather than storing navigation in SQL
src/config/builderUx.ts	builderUx feature flags	H	No	Visibility, NOT product removal	Feature keys	Jamb material/color and glass-frame color controls currently hidden; underlying data remains
src/utils/matchingFrameFinish.ts	matchingJambFinish	H algorithm	No	Shared finish-ID match; first available clad fallback	IDs	Matching hidden jamb finish when door color changes
src/types.ts	DoorStyle, DoorStyleVariant, DoorLine, Finish, GlassOption, hardware types, DoorConfiguration, grid/sidelite/coating/swing unions	H contracts	Asset URL fields	Supported value domains, not live inventory	Both	Boundary contracts throughout app/server; do not remove historic fields

Consumer and secondary-rule sources

File	Role and inventory relevance
src/components/DoorPreview.tsx	Applies resolved slab, glass/frame, sidelite, finish and hardware data. FINISH_RENDERING and layered masks/tinting are material rendering behavior, not alternate color inventory.
src/features/home-visualizer/entranceGeometry.ts	ENTRANCE_GEOMETRY, STANDARD_SIDE_LITE_TO_SLAB_RATIO, createCanonicalEntranceGeometry: fixed render coordinates and proportions shared by preview/PDF/Visualizer.
src/features/home-visualizer/captureDoorPreview.ts	Captures the configured entrance; consumes resolved product visuals, does not supply inventory.
src/features/home-visualizer/entranceFitStrategy.ts	getDetectedVisualizerOpeningFamily, getSelectedVisualizerOpeningFamily, evaluateEntranceCompatibility: photo-family guidance/blocking, NOT manufacturer product compatibility.
server/aiDoorVisualization.ts	sideliteCatalogs, resolveAiProduct, asset/reference resolution: imports product data but repeats some classification/fallback logic and resolves sidelite glass by name. Server/browser must use the same catalog revision later.
src/utils/doorBuilderDraft.ts	Version-1 draft sanitization/restoration validates IDs against current catalogs. Disabling a product later must distinguish new selection eligibility from historic renderability.
src/utils/pdfProductRenderer.ts	Consumes shared assets/colors/hardware in the deterministic PDF preview. PDF render canvas dimensions are not manufacturing sizes.
src/utils/pdf.ts, src/utils/pdfConfig.ts	Consume configuration labels and fields; PDF template/placement/download are not catalog sources.
src/utils/submission.ts, api/submit-door-builder.ts	Translate configuration into existing submission fields and Supabase leads/Zapier flow. Preserve payload contracts and historic values.
src/utils/sortByDisplayLabel.ts, src/styles.css	Display ordering, responsive layout, preview sizing. No inventory migration required.
server/dealerResolution.ts, api/dealer-context.ts	Existing Supabase dealers lookup; not inventory.
server/aiUsage.ts, api/complete-ai-visualization.ts	Existing Supabase ai_generation_usage / ai_entrance_detection_usage telemetry; keep separate.
public/assets/**, asset audit/optimization/hero scripts	Actual deployed art, path integrity and generated marketing assets. A physical asset is not necessarily a customer product. Do not run optimization as an inventory migration.

3. Categories and counts

Counts are from runtime imports of the current source modules, source inspection and the existing read-only asset/integrity checks. They are **not stock counts or independent SKUs**. Parent products, material variants, finish-specific variants, raw rows, visible rows, and asset expansions are deliberately distinguished.

Additional categories represented in the code include glass coming/metal variants (for example Nickel/Patina/Brass within named designs), mini-blind variants, CLiC glazing/grid options, Low-E coatings, double-door preparation and Savannah hinge options, timber versus clad frame context, glass-frame finishes, interior/exterior artwork views, and marketing presets. These are not all independent stocked product masters. A transom is currently an analyzed photo feature rather than a customer-selected transom inventory catalog.

Category	Count	Meaning
Customer-facing door line/material groups	5	Signature Series Fiberglass Grained, Smooth Steel, Paintable and Stainable Steel, Brushed Smooth Fiberglass, Textured Fiberglass
Internal catalog line records	8	Four Signature grain lines plus four other lines
Broad base materials	2	Steel and fiberglass, inferred from line names; no independent material master table
Explicit steel gauge-coded internal lines	2	20-gauge-smooth-steel, 22-gauge-steel; gauge is not a separate picker and is absent from customer line labels
Named grains	4	Cherry, Fir, Mahogany, Oak; Oak also used by other textured/stainable lines
Door style/design records	42	catalogDoorStyles, grouped across line variants
Style-line variants	127	Sum of line style rows; 43 distinct raw variant codes
Entry configurations	3	Single, French, Savannah
Handing/swing choices	4	LHI, LHO, RHI, RHO
Double-door lock-prep choices	3	DDLLBO, DDLLAC, DDLLKP
Savannah required product option	1	HINGEOJ; behavior, not a standalone stocked product list
Paint colors	26	paintColors
Stain colors	21	stainColors
Combined finish records	47	options.finishes; not 47 valid finishes for every line

Category	Count	Meaning
Clad colors	6	Restricted subset of paints, not six additional colors
Main glass families	5 + conditional Retro	Clear, Decorative, Privacy, Mini Blinds, CLiC; Retro substitutes for family grouping on specified styles
Approved decorative base names / IDs	23 / 35	Approval allowlists; not all imported source rows
Approved privacy base names / IDs	6 / 8	Includes Linen in approval layer, separately hidden in customer layer
Main glass exported rows / unique IDs	127 / 126	Duplicate par is row
Main glass rows passing customer-only Linen predicate	126	Still subject to style/material/legacy filtering
Main glass IDs reachable through current material/style and visibility filters	88	Static enumeration of current line choices and styles, reproducing availableGlass, additive allowlist updates and Linen predicate; not a count for any single door
Main glass IDs retained but not reachable by that enumeration	38	Listed in section 7; no discontinuation claim
Sidelite slab styles	5	FSL, F48SL, SSL, S2SL, CR14SL
Sidelite asset families / family-style associations	7 / 21	Family aliases can share the same physical artwork
Semantic sidelite relationships / physical codes	4 / 4	None/hinge-side/lock-side/both-sides; NOSIDE/LEFTSIDE/RIGHTSIDE/BOTHSIDES
Exported sidelite glass rows	92	FSL 31, F48SL 21, SSL 29, S2SL 1, CR14SL 10; scoped rows, not 92 globally unique designs
Sidelite glass categories by slab	5 / 3 / 4 / 1 / 3	FSL / F48SL / SSL / S2SL / CR14SL
Hardware brands	2	Baldwin, Schlage
Hardware manufacturer/style groups	15	Grouped picker products
Hardware finish-specific selectable records	34	Derived hardwareOptions
Actual hardware finishes	5	Bright Brass, Dark Bronze, Matte Black, Satin Nickel, Venetian Bronze
Baldwin / Schlage source records	13 / 21	Different source representations; not view-expanded assets
Derived hardware asset records	97	Includes view/handing expansions; NOT 97 selectable hardware products
Jamb material choices retained internally	2	Timber, clad; UI currently hides material/color selection

Category	Count	Meaning
Finish kinds	2	Paint/stain; clad is a frame finish context/subset, not a third slab finish palette
Grid location / profile domains	5 / 4	Type unions: No Grids/External/Internal/SDL/Arts & Crafts and Arts & Crafts/Contoured/Flat/Prairie; UI has a smaller scoped set
Grid pattern / color / width domains	11 / 7 / 3	Numeric lite patterns including horizontal variant; widths 5/8, 7/8, 11/16 inch; not every cross-product is valid
Coating label domain	7	Includes composite “either coating” labels; not seven independently selectable coatings everywhere
Main glass opening masks	33	Code -> mask entries
Default smooth slab map / thumbnail map keys	39 / 44	Aliases and auxiliary keys mean neither equals visible style count
Sidelite glass masks	5	One per slab style
Hero presets	29	Marketing combinations, not inventory

Line/style membership totals

Internal line	Style rows	Grain	Customer group
signature-cherry	9	Cherry	Signature Series
signature-fir	2	Fir	Signature Series
signature-mahogany	5	Mahogany	Signature Series
signature-oak	3	Oak	Signature Series
20-gauge-smooth-steel	37	None	Smooth Steel
22-gauge-steel	33	Oak	Paintable and Stainable Steel
brushed-smooth-fiberglass	24	None	Brushed Smooth Fiberglass
textured-fiberglass	14	Oak	Textured Fiberglass

Do not rename gauge-coded IDs during import merely because customer labels have changed. Existing drafts and integrations depend on the IDs.

Asset counts and limitations

scripts/auditProductionAssets.ts scanned 91 source files and found:

- 1,572 raster files under deployed assets, totaling 33,232,008 bytes.
- 716 exact source reference strings and 18 dynamic prefixes.
- 1,571 conservatively referenced files after prefix expansion; 33,135,636 bytes.
- Zero missing exact references.
- 63 byte-identical duplicate groups containing 79 additional copies.
- One candidate unreferenced raster: public/assets/visualizer/auto-fit-door-slab-example.webp (96,372 bytes).

These totals include marketing, UI, PDF and Visualizer imagery, not only inventory art. Dynamic-prefix expansion intentionally overcounts reachability; it cannot prove that all 1,571 files are used by a currently selectable product. Conversely, SVG/generated procedural images and arbitrary future remote assets are not represented by this raster count. A later import should produce an explicit product-role asset manifest before claiming an exact inventory-only asset count.

The configuration integrity test separately checked 786 unique asset paths and 357 unique AI reference sets across 50,381 prepared AI configurations, with no missing assets. Its 51,081,189,421,440 theoretical combination estimate is an automated test estimate, NOT a manufacturer SKU count, stock count, or guarantee of commercial availability; it multiplies derived rule domains and includes retained records.

4. Relationships and current availability rules

Door styles, lines and materials

productCatalog establishes explicit style membership per internal line. catalogDoorStyles merges rows by normalized **display name**, then stores variants[] containing line ID, line name, original style code, grains and color allowance. The visible line group can contain several internal lines. doorLineChoicesForStyle selects groups whose lineIds intersect the style variants.

s1NGSS is displayed as S1 / Six Panel No Glass, which merges it with S1 variants. Preserve both raw variant codes and the existing stable derived style ID. Future names must not become primary keys: a display-name edit currently risks changing the derived style slug/grouping.

glassDoorCodes is a 29-code allowlist; doorStyleSupportsGlass checks selected-line variants. A shared style's aggregate hasGlass alone is not sufficient to choose a line-specific glass path.

Finishes and colors

finishesForStyle/finishTypesForDoorLine combine variant membership, allowed grains, color allowance, paintOnly and line grouping. Smooth Steel and Brushed Smooth Fiberglass are paint-only. Textured Fiberglass and Paintable/Stainable Steel use Oak automatically; Signature chooses compatible grain variants.

App narrows the available finish records by active paint/stain kind and resolves frame choices separately. Hidden selectors do not remove those fields: jamb/glass-frame values still flow into the preview, configured reference, PDF and submission. Timber can reuse the door finish ID. Clad matching uses the same ID when available, otherwise the first clad finish. Glass frame has match-door/custom modes and a compatible default (paint-white or first available). Do not silently expand clad availability to all paints/stains in the migration.

Main glass

There is no clean main-glass product-to-family foreign key today. Family is inferred from ID/name containing CLiC/blinds, clearGlassIds, privacy approval membership, then decorative fallback. Retro is a presentation grouping for 3LT/3STEP/4LT/5LT/F764, not another independent glass inventory.

Actual availability is a sequence:

- 1. Build rows from thumbnail/variant source arrays; discard umbrella/legacy thumbnail entries as encoded in glassoptions.ts.
- 2. Apply known/approved privacy and decorative ID allowlists.
- 3. Require an overlay for at least one selected style/line code.
- 4. Exclude legacy clear/blinds sets on F, S and F48/F482; enforce F48 whitelist (including its later .add entries).
- 5. Exclude privacy glass for SAT.
- 6. Hide Linen with isCustomerSelectableGlass at picker presentation.
- 7. Group named variants and sort visible choices.

An overlay being available is currently both an asset fact and an availability signal. Database relationships should explicitly record allowed membership instead of inferring all commercial compatibility from asset existence.

Sidelites

sideLiteAssetFamilyForSlab maps line/grain to family; map keys define allowed slab styles:

Asset family	Allowed sidelite styles	Current line context
smooth-steel	FSL, F48SL, SSL, S2SL	Smooth Steel
20-gauge	FSL, F48SL, S2SL	Brushed Smooth Fiberglass
22-gauge	FSL, F48SL, SSL, S2SL	Paintable/Stainable Steel and Textured Fiberglass
cherry	FSL, F48SL, SSL	Signature Cherry

Asset family	Allowed sidelite styles	Current line context
fir	FSL, CR14SL	Signature Fir
mahogany	FSL, F48SL, SSL	Signature Mahogany
oak	F48SL, SSL	Signature Oak

These are current asset-family aliases, not proof that fiberglass is a steel gauge. Smooth Steel SSL intentionally reuses `sidelites/Signature/Oak/SSL.webp`; do not exclude it based on filename/gauge. `doorStyleCode` exists in the family input type but the current family resolver uses line/grain, not a style-by-style sidelite manufacturer table.

Each slab chooses its own glass catalog and grid rules. CR14SL derives glass from main CR14 overlays; S2SL has one clear option and no populated grid rules. Semantic hinge-side/lock-side is resolved using handing and exterior/interior view; French/Savannah can present physical left/right options translated through the same code mapping. This translation is application behavior, not four independent sidelite products per door.

Hardware

Source artwork expands into view/handling assets. `hardwareOptions` selects exterior/preferred handing records, deduplicates by its derived key and groups manufacturer/style with finish choices. The inspected path does not contain an independent comprehensive door-style/material -> hardware commercial exclusion table. Do not invent one or assume all source asset rows are distinct sellable products.

`doorHardwarePlacements` applies selected preparation to double doors: DDLLBO full locksets on both leaves; DDLLAC full lockset on active leaf; DDLLKP full active lockset plus knob-only inactive hardware. Savannah retains HINGEOJ behavior. These are precise rendering/functional semantics; importing labels alone would lose them. Single-door handing is unchanged by finish/style grouping. Temporary Georgian/Matte Black preview art before selection is a UI fallback, not a saved default hardware order.

Grids, coatings and sizing

Main F, F48/F482 and S glass each have scoped standard/Low-E maps. The key sequence is glass opening -> grid location -> profile -> numeric pattern -> color -> width, with coating inferred from standard/Low-E rule intersection. F48 removes external grids. External/SDL patterns have explicit opening-specific lists. Sidelite grids use separate slab-specific maps, with shared Prairie/Arts & Crafts art and fallback choices. Empty width arrays mean a widthless/no-width-selection case; `undefined` means disallowed. Do not conflate them during import.

Type unions contain 11 patterns and seven colors; these are domains, NOT all-pairs availability. Generated paths use different codes by opening (BK VS BL, BZ VS BR). Bronze/White can reuse White artwork with distinct rendering. Import exact resolved assets per valid grid tuple, rather than guessing filenames.

No customer manufacturing width/height selector was identified. Named lite dimensions such as F848 and S836 and grid bar widths are product descriptors. `ENTRANCE_GEOMETRY` uses slab width 242, opening height 549, sidelite ratio 0.35, center stile 7, threshold 12 and frame profiles in render coordinates. The comment references a 36-inch slab/12-14-inch sidelite as a fallback convention; these numbers are NOT an approved manufacturing size catalog. CSS and PDF canvas sizes likewise stay out of physical inventory dimensions.

Options disappearing and configuration consumers

App derives `available*` lists and clears invalid downstream selections when style/line/grain changes. It skips unsupported glass/grid pages and hidden jamb/frame-color controls. Default selection and fallback chains rely on array order in some places, while `sortByDisplayLabel` sorts only rendered copies; database `display_order` must not inadvertently change defaults.

Existing drafts store IDs and selected values; some downstream configuration fields additionally carry nested product data and labels. Sidelite glass is translated into a name/asset configuration and server lookup can use name equality. PDF and leads depend on these existing contracts, so an adapter must preserve them even if database records use new UUID keys.

Visualizer A-E photo-opening compatibility is a separate concern: A single; B single + one sidelite; C single + both sidelites or double without sidelites; D double + one; E double + both. This is not stock availability and must remain in application code during inventory migration. Entrance detection/model/prompts/masks are out of scope.

5. Asset mapping system

Assets live in `public/assets` and are served through `/assets/...` paths. Major inventory-related branches include `door-lines`, `door-styles`, `finishes/paint`, `finishes/stain`, `glass/thumbnails`, `grid-options`, `hardware`, `hardware/cards`, `hardware/finishes`, `hardware/lock-setup`, `masks`, and `hgi-assets` (Preview Slabs, Preview Hardware, glass opening families and Sidelites). `generated/hero-doors` is marketing output; `pdf-icons` is document UI art.

The same product can need several roles: picker thumbnail, slab render, finish swatch, glass overlay, glass opening mask, frame mask, hardware card, exterior/interior hardware, grid overlay and sidelite slab. One `image_url` column per product would be insufficient.

Slab candidates are selected by code + line + grain + finish context, then fallbacks. Brushed Smooth Fiberglass deliberately returns only smooth candidates so missing artwork cannot silently turn it into textured/grained material. Other lines can fall through alternate maps and thumbnails. Preserve precedence and log fallback use; do not replace this with “first asset for style.”

Main glass overlays are keyed by opening code. SO2 can reuse SO art. FSL/F48SL/SSL construct paths by codes; several grid/hardware products intentionally share art. Hardware has both card art and installation art with view/handing metadata (`?v=5` is URL cache-busting, not a product version). Glass-frame geometry is procedurally derived and should not become admin-editable SQL expressions.

Recommended asset manifest fields: stable asset ID, existing public path, role, MIME, width/height, byte size, checksum, transparency, source/provenance, content version and optional crop metadata. Store per-product binding context separately. Retain current local paths in the first migration; moving media to Supabase Storage is a separate reversible phase.

6. Product facts versus executable rules

Suitable for database administration: display names/descriptions, internal codes, series/material/grain identity, paint/stain hex palettes, explicit style-line membership, explicit allowed finish/glass/sidelite/grid tuples, customer visibility, active/legacy/discontinued status, asset metadata/bindings, notes, approved ordering and manufacturer provenance.

Keep in code: step/progress/navigation, selected-state effects, sorting implementation, semantic sidelite/handing/view translation, active-leaf hardware placement and knob crop semantics, rendering geometry/tinting/mask creation, complete reference capture, PDF layout, submission serialization, draft validation/migration algorithms, photo normalization/detection, A-E compatibility, automatic Good-fit continuation, generation/model/prompt/error/telemetry behavior.

Hybrid examples: store supported grid combinations but keep the evaluation/defaulting algorithm in code; store approved glass-family membership but keep presentation grouping; store material traits such as paint-only but keep validation of the resulting selection. Do not introduce a general-purpose rule scripting engine in Supabase.

7. Problems and inconsistencies found (not fixed)

1. **Duplicate main glass ID:** `paris` occurs in both thumbnail sources and survives as two exported rows. Both currently have the same label, thumbnail and eight opening keys. A future unique constraint/import needs an explicit deduplication decision and legacy-ID mapping, not silent last-write-wins. No duplicate IDs were found in the 42 derived door styles, 47 finishes, or 34 selectable hardware rows.

2. **Duplicate labels with different scope:** `Grace-Nickel/Patina`, `Nouveau-Nickel/Patina`, `Clear Glass`, `Clear Glass with Grids`, `NOGRID` and `NONSTOCKCL` labels recur under different IDs/opening families. Examples: `ca-grace-nickel/grace-nickel`, `hrt-nouveau-nickel/nouveau-nickel`, and `f-clear-grids/f48-clear-grids/s-clear-grids`. These are not necessarily duplicates; scope and raw codes must survive import.

3. **Retained but currently unreachable main glass IDs:** static enumeration found 38:

```
contg, decorative, extg, flatg, linen, ocean-caming, privacy, celestial,
blinds-espresso, blinds-gray, blinds-sand, blinds-silver, blinds-tan, blinds-white,
f-f10l, f-f15wh, f-prairie-internal, f-blinds-15, f-ten-lite,
f-clear-f10, f-clear-f10l, f-clear-f15, f-clear-f15int, f-clear-f15intl,
f-clear-fpaint, f-clear-ften, f-clear-nonstock,
f48-clear-f1248, f48-clear-f1248l, f48-clear-f648l, f48-clear-nonstock,
s-clear-s5, s-clear-s5l, s-clear-s9, s-clear-s9int, s-clear-s9intl,
s-clear-sv6, s-clear-nonstock
```

Includes placeholders, explicit legacy exclusions, lack of matching overlay/membership and `Linen`. This is not proof of discontinued manufacturer products. `F48` .add updates were included; omitting them incorrectly understates visible IDs by two. Source approval filters also remove other raw designs before export; preserve that distinction when extracting source tables.

4. **Linen policy layering:** privacy approval still includes `Linen` while customer predicate hides it in main and `FSL/F48SL/SSL` pickers. Model `customer_selectable` separately from `renderable/legacy` availability; do not delete saved values.

5. **Display-name identity coupling:** styles merge and get IDs from normalized names; hardware IDs derive from manufacturer/style/finish/handing labels. Label changes can break historic IDs. Assign stable database identities and preserve legacy external IDs.

6. **Gauge/family names are not material truth:** customer labels omit steel gauge; filenames/internal IDs retain gauge names. Brushed fiberglass uses a 20-gauge sidelite asset alias, textured fiberglass a 22-gauge alias. Never infer sellable gauge/material from file path.

7. **Availability lives in multiple places:** catalog flags, asset existence, approvals, App whitelists/exclusions and hidden-control flags all contribute. A source-array-only import would expose hidden/unsupported choices or remove valid ones.
8. **Shared assets and duplicate bytes:** 63 identical groups; examples include Schlage Bright Brass interior art shared across authored/card-style paths, multiple Baldwin style previews sharing finish/view art, and S-family clear-grid overlays (SF10, SF5, SF5L, SS5L). Smooth Steel SSL shares Oak SSL artwork; F48SL/SSL Arts & Crafts share one path. Shared art can be intentional; checksum identity does not imply product identity.
9. **Mapping-only hardware finish:** `finishColors` includes Aged Bronze, but current selectable hardware rows use five other finishes. Do not expose it solely because a color map exists.
10. **Candidate stale asset:** `visualizer/auto-fit-door-slab-example.webp` is the audit's only unreferenced raster candidate. Not deleted. Broad dynamic prefixes may conceal other unused files.
11. **Fallbacks can hide fidelity gaps:** available assets may render through alternate slab maps/thumbnail fallback; geometric rules use generic defaults. Zero missing files does not prove correct artwork for every commercially valid combination. Need per-role visual acceptance during migration.
12. **Server/browser drift risk:** shared imports coexist with repeated resolver logic and name-based sidelite lookup. Database switching must be revision-consistent in both layers; cannot migrate browser lists alone.
13. **No authoritative stale-product determination:** current code has no discontinuation dates or live manufacturer inventory feed. Legacy/hidden records are candidates for HGI review, not products to remove automatically.
14. **No physical sizing/stock source:** grid widths and render pixels are not orderable door dimensions. Manufacturer availability disclaimer remains necessary. Confirm actual manufacturing constraints with HGI before adding dimensions, quantities or commercial promises.

8. Proposed Supabase schema (design only)

Prefer a catalog schema with stable UUID primary keys and explicit foreign keys. Preserve current IDs/codes through unique legacy-key columns scoped to the correct namespace. Do not make customer names or asset filenames primary keys. No SQL was applied.

Common lifecycle fields

Customer-editable product/option tables should have `legacy_id`, `internal_code` where applicable, `display_name`, `description`, `status` (active, legacy_only, discontinued), `customer_selectable`, `display_order`, `notes`, `created_at`, `updated_at`, and `publish/audit provenance`. `customer_selectable=false` must not prevent resolving historical saved configurations. Junction rows can carry status/order/notes when that relationship has its own lifecycle. Unique keys are scoped (e.g. slab + sidelite glass legacy ID), not globally enforced on repeated labels.

Display order is explicit editorial data; the application retains its current alphabetical/semantic sorting policy and default selection priority. Store a separate `default_priority` only where needed to preserve existing first-option fallbacks.

Tables and why they exist

Proposed table(s)	Main fields / relationships	Purpose
catalog_revisions	ID, status draft/published/retired, published_at/by, manifest checksum, source commit	Publish a validated complete snapshot, not half-edited live arrays
materials, series, grains	Stable codes/names; steel/fiberglass traits; series -> material; grain metadata	Separate commercial identity from asset aliases/gauge filenames
door_lines	Series/material FK, optional gauge metadata, paint/stain traits, customer group reference	Preserve eight internal line records and five current groups without adding a gauge UI
door_line_groups, door_line_group_members	Customer group name/code/order; group↔line FKs	Model grouped Signature choices explicitly
door_styles, door_variants	Style FK + line FK + original product code, grain association, glass-opening code/has-glass	Represent 42 shared designs and 127 line memberships without name-based merging
finishes, door_variant_finishes	Kind paint/stain, hex, ProMatch; variant↔finish allowlist	Reuse 47 finishes and explicitly express allowed combinations
frame_materials, frame_material_finishes	Timber/clad records; allowed finish FKs and fallback priority	Preserve frame palette/subset independently of hidden selectors
glass_families, glass_designs, glass_options	Family/design FKs; scoped legacy ID, raw code, variant/caming/coating metadata	Distinguish a design such as Grace from finish/opening-specific options; no name parsing required
door_variant_glass	Variant FK + glass option FK + opening code; availability/status	Explicit main-glass compatibility, independent of overlay presence
sidelite_styles, sidelite_variants, door_variant_sidelites	Style, line/grain variant, original code; allowed door variant↔sidelite variant	Separate five sidelite designs from seven art families; retain SSL eligibility
sidelite_variant_glass	Sidelite variant FK + scoped glass option FK	Scoped sidelite glass compatibility; preserve CR14-derived membership at import
hardware_brands, hardware_styles, hardware_finishes, hardware_variants	Brand/style/finish FKs; legacy selected ID, functional kind and knob-prep metadata	15 grouped styles and 34 sellable finish records; artwork view/handling not fake products
entry_product_options	Codes/labels for retained order-facing prep/hinge options	Optional small catalog of DDLL* / HINGEOJ metadata; their execution remains in code
grid_profiles, grid_patterns, grid_colors, grid_widths, glass_coatings	Stable codes/labels, physical grid widths, numeric order	Small typed dimensions reused by main/sidelite grid combinations

Proposed table(s)	Main fields / relationships	Purpose
grid_combinations plus door_glass_grid_combinations / sidelite_glass_grid_combinations	Opening family, coating, location, profile/pattern/color/width nullable where meaningful; typed allowed-membership FKs	Import allowed tuples rather than unrestricted Cartesian products; preserve widthless vs forbidden cases
assets	Storage/public path, role-neutral metadata, MIME/dimensions/bytes/checksum/version/provenance	One media record can serve several products without duplicated bytes
Typed asset-binding tables: door_variant_assets, glass_option_assets, sidelite_variant_assets, hardware_variant_assets, finish_assets, grid_combination_assets, plus line/group card bindings	Parent FK + asset FK + role + opening/grain/finish/view/handing context and candidate priority as applicable	Supports thumbnail/render/mask/view art with referential integrity; avoid unvalidated polymorphic product IDs
catalog_change_log	Editor, timestamp, revision, entity and before/after metadata	Future admin audit trail; separate from customer/lead records

The above is a logical table proposal, not a mandate to build all admin screens at once. Small immutable domains such as entry type/handing may remain application enums initially. Do not precompute/store tens of trillions of full door configurations. Join small catalogs and scoped allowlists at runtime.

Use stable entities with revision-scoped published memberships/values or immutable published snapshot manifests so one builder session can remain pinned to a revision. Choose that concrete storage approach before writing migrations; a revision row by itself does not version mutable product rows. Existing submission JSON should keep the selected names/codes and ideally later include catalog revision plus stable IDs, preserving historical quotes when catalog names change.

Only add hardware-door commercial allowlists after HGI supplies authoritative restrictions; current code does not justify invented restrictions. Physical manufacturing size_options and stock tables should be a separate later extension after units, stock scope and supplier semantics are agreed.

Security and administration

Expose only published customer-selectable catalog reads; restrict draft edits/publication to authenticated authorized admins. Keep notes intended for staff out of anonymous responses. Use RLS and explicit privileges on exposed tables, and never ship service-role credentials to the browser. Existing lead/dealer/telemetry access must remain unchanged. See [Supabase RLS guidance](#).

If media moves to Supabase Storage later, restrict write/delete through storage policies and authorized admin roles. Published product art can be public without exposing customer photos. See [Supabase Storage access control](#). Inventory admin editing should not use the existing customer lead endpoints.

9. Migration safety plan

1. **Freeze and export a baseline:** preserve this source commit, scoped legacy IDs, all source and effective visible memberships, default priorities, approvals/exclusions and explicit resolved asset manifests. Include hidden/legacy records and why they are hidden. Resolve duplicate Paris ownership with HGI before applying uniqueness constraints.

2. **Build Supabase alongside current code:** new tables/admin publication, no production reads switched. Keep existing assets local initially. No FeneVision dependency.
3. **Import idempotently:** upsert by scoped stable legacy key; stage raw and effective records separately. Record source provenance and counts. Do not infer material from paths, merge by label, or mark hidden records discontinued automatically.
4. **Introduce an adapter in a later implementation phase:** return exactly the existing TypeScript shapes from either source, validating types, IDs, relationships and asset roles. Keep the hardcoded adapter as the reference. This phase does not add that adapter.
5. **Shadow comparison:** fetch Supabase inventory without driving customer UI. Diff IDs, labels, code membership, approved/hidden options, finish hex/type, render candidates and priority, resolved assets, defaults, and grid tuples against hardcoded output. Test loaded drafts and nested submission/PDF values, not counts alone.
6. **Migrate one low-risk category at a time:** suggested sequence palettes -> hardware catalog -> door lines/styles -> main glass -> sidelites -> grid compatibility. Each category requires complete dependent memberships and asset bindings before switch. Keep behavioral algorithms unchanged. Treat interdependent glass/overlay changes atomically.
7. **Feature flags and coherent fallback:** opt-in staging/canary first; pin a builder session to one validated published catalog revision. Use bounded fetch timeouts, last-known-good snapshot and whole-snapshot/category-boundary fallback, never half remote/half local relationships after selection. Log source/revision/fallback without customer images or secrets.
8. **Historic draft protection:** distinguish selectable from renderable records; load a historic product with its saved revision/legacy identity where supported. If unavailable, explain and request reselection rather than silently substituting another product. Preserve current draft schema/version safeguards until an explicit migration is tested.
9. **Parity gates:** current integrity/browser/draft/handling/PDF/submission tests; direct member-set comparison; preview/reference/PDF visual samples; mobile; valid/invalid commercial combinations; disabled-product history; duplicate import; missing asset; network outage; admin publish rollback and server/browser revision mismatch.
10. **Production validation before removal:** monitor product resolution/fallback rates, test real orders with HGI, restore previous published snapshot via flag/revision if needed. Remove hardcoded runtime inventory only after a documented validation period; retain export snapshots/legacy aliases for rollback/history.

Acceptance means the same configurations remain selectable and render identically, existing saved values/PDF/submission fields remain valid, and catalog updates require admin publication rather than application redeployment. It does not mean changing product rules or the Visualizer during migration.

10. Risks, recommendations and verification

Highest risks: label-derived IDs; hidden records accidentally exposed; asset existence mistaken for commercial approval; different source revisions in browser/server; silently invalidating saved drafts; losing grid null/empty semantics; material inferred from gauge-named asset directories; changing defaults through DB ordering; editing colors that alter renderer fidelity; and coupling catalog downtime to normal builder access.

Admin editing can add products without redeployment only within the renderer's supported data/asset contract. A genuinely new opening shape, material rendering method or functional hardware mode may still need code support; do not promise arbitrary new geometry through inventory tables alone.

Before implementation, ask HGI to confirm commercial availability versus physical stock scope, stable manufacturer codes, hidden versus discontinued status, ownership of duplicate/legacy glass records, acceptable fallback colors, actual manufacturing dimensions, and approval responsibility. Keep admin changes draft until all assets/memberships validate and an authorized person publishes a coherent revision.

Checks run for this report

Check	Result
npm run test:configuration-integrity	Passed: 4,688 material/finish selections, 786 asset paths, 50,381 AI product preparations, 357 reference sets; zero missing/broken/deleted-raster references
node --import tsx scripts/verifyCustomerGlassAvailability.ts	Passed: Linen hidden, retained catalog unchanged; main 126 other rows, FSL 30, F48SL 20, SSL 28
npm run test:door-builder-draft	Passed persistence and defensive restoration checks
npm run test:entrance-compatibility	Passed 64 A-E structure combinations and confidence/transom safeguards
npm run test:option-order	Passed 1280px desktop and 390px mobile F/F1 flows; display ordering, filtering, saved selections, keyboard focus, no overflow
node --import tsx scripts/verifySideliteHanding.ts	Passed 32 relationship/handing/view mappings plus preview/capture/PDF/AI/submission/legacy-draft checks
node --import tsx scripts/auditProductionAssets.ts	Read-only audit passed with zero missing exact references; duplicate/unreferenced candidates above
npm run build	Passed TypeScript project build and production Vite build; existing large-chunk warning (notably HEIC bundle), not a build failure
git diff --check	Passed; separate git diff --no-index --check /dev/null docs/inventory/inventory-audit.md also produced no whitespace errors for the new untracked report (exit 1 indicates file differences)

No live OpenAI, production Supabase, admin publication, lead submission or paid generation request was made. Counts/relationships were inspected locally; manufacturing stock accuracy cannot be verified from this repository. The requested deliverable is this report only; no commit is made automatically.